

EMMANUEL CHUKWUEBUKA MOGHALU

Data Engineer | Backend Systems Engineer

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PROFESSIONAL SUMMARY

Data and backend engineer with 4+ years of experience across full-time, part-time, contract, and freelance roles, building production-grade APIs, event-driven data pipelines, and analytics platforms — validated with automated test suites, not just design documents. Deep working knowledge of the Nigerian and broader African fintech ecosystem, including multi-PSP integration (Paystack, Flutterwave, M-Pesa, Moniepoint, OPay, Interswitch), CBN regulatory requirements, and NDPR compliance. Comfortable owning a system end-to-end — from PRD and data architecture through implementation, testing, and dashboarding — and communicating technical tradeoffs clearly to cross-functional stakeholders.

CORE TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Backend Engineering: Django, Django Ninja, FastAPI, NestJS, REST APIs, Next.js/React, Microservices, API Design, Authentication, SSE/WebSockets

Data Engineering: dbt, Dagster, DuckDB, Apache Spark, PostgreSQL, Alembic, ETL/ELT Pipelines, Data Modeling, Data Warehousing, pgvector, Semantic Search / Embeddings

Event-Driven & Streaming: Kafka/Redpanda, Celery, Prefect, Redis

Analytics & BI: Pandas, Streamlit, Grafana, Power BI, Plotly, Prometheus, KPI Development, Dashboarding

Storage: PostgreSQL, MinIO (S3-compatible object storage), Redis

DevOps & Infrastructure: Docker, Docker Compose, Git, GitHub Actions, CI/CD, Linux, structlog, Monitoring & Logging

Domain Expertise: Nigerian/African Fintech (CBN & NDPR compliance, PSP integration), System Design, Distributed Systems

PROFESSIONAL EXPERIENCE

Software & Data Engineer (Contract)

Jan 2026 – Feb 2026

MedVax Health — E-Pharmacy / Telemedicine Platform

- Architected a secure e-prescription portal supporting prescription management, authentication, and payment processing workflows for doctors and patients.
- Optimized data flow and state management for real-time updates, improving user engagement across the platform.
- Implemented strict validation and error handling, improving system reliability and application stability; investigated production issues via structured debugging and root cause analysis.

Technologies: Python, FastAPI, NestJS, Prisma ORM, TypeScript, PostgreSQL, REST APIs, Docker

Data Operations & Systems Engineer

Mar 2024 – Oct 2024

SETRACO Nigeria Ltd (completed alongside final year of B.Eng.)

- Designed and implemented centralized reporting solutions that replaced manual Excel-based workflows, reducing reporting effort by approximately 40%.
- Built SQL-powered dashboards that improved operational visibility and accelerated management decision-making.
- Maintained 99.9% data accuracy across production reporting systems through data validation and quality assurance.
- Diagnosed and resolved SQL pipeline failures and data inconsistencies, improving overall operational reliability.
- Optimized SQL queries and backend processes to improve reporting performance and scalability.

Technologies: Python, SQL, React (Next.js), PostgreSQL, Dashboard Development, Data Analysis

Freelance Software & Data Engineer

Oct 2020 – Mar 2024

Self-Employed (concurrent with B.Eng., Caritas University)

- Delivered web development and data solutions for independent and small-business clients, managing full project lifecycles from proposal through delivery.
- Authored client-facing technical proposals and scoped project requirements.
- Designed and shipped a personal portfolio site (React, TypeScript, Tailwind CSS, Framer Motion).
- Sustained continuous hands-on engineering work throughout a full-time engineering degree program.

Technologies: Python, JavaScript/TypeScript, React, SQL

Full-Stack Software Engineer (Part-Time)

Aug 2019 – Aug 2020

TAC AFRICA (concurrent with AfriHUB ICT Solutions, below)

- Developed and maintained web applications serving over 50,000 users, contributing to improved platform stability and user experience.
- Optimized backend APIs and SQL queries, reducing response latency by approximately 45%.
- Investigated and resolved production issues through structured debugging, log analysis, and database optimization.
- Collaborated with designers and product teams to implement new features and improve application performance.

Technologies: Python, JavaScript, SQL, REST APIs, PostgreSQL

Software Engineer

Oct 2018 – Oct 2020

AfriHUB ICT Solutions (undertaken alongside the National Innovative Diploma program, below)

- Designed and implemented software solutions for academic and business clients using modern web technologies.

- Diagnosed and resolved application, database, and integration issues in production environments.
- Supported deployment, system integration, and maintenance of enterprise applications.
- Assisted in building secure database structures and backend business logic for client projects.

Technologies: Python, SQL, HTML, CSS, JavaScript, MySQL

SELECTED PROJECTS — BUILT & DEPLOYED

Cross-Border Mobile Money Reconciliation Engine

- Built a payment reconciliation platform for Nigerian multi-PSP operations (Paystack, Flutterwave, M-Pesa, Moniepoint, OPay, Interswitch, Remita, Paga), targeting an industry problem where finance teams lose 30–40% of their time to manual reconciliation.
- Engineered a Medallion pipeline — FastAPI ingestion, Redpanda (Kafka-compatible) event queue, MinIO/Parquet Bronze layer, PostgreSQL Silver layer (PII-masked, FX-normalised), dbt Gold layer — with HMAC-authenticated webhooks, scheduled polling fallback, and automated gap detection.
- Built a two-tier matching engine (exact + trigram-based probabilistic) with 5-category discrepancy detection, engineered toward a 99.5% match-rate target with sub-10-second end-to-end latency and exactly-once processing semantics.
- Shipped with 160+ automated tests across 9 suites, database-enforced PII masking, an immutable audit trail, and a 7-screen Next.js/React executive dashboard with CBN-format reporting.

Repo: github.com/emmanuelrichard01/mmr-engine

ULTRA-NEWS V3 — The Wire Room (News Aggregation Platform)

- Rebuilt a legacy Django monolith into a story-centric aggregation platform that clusters articles from 35+ RSS sources into verified Story objects through a three-tier lifecycle (Wire → Developing → Reporting).
- Replaced exact-title-match clustering with semantic vector clustering (pgvector + local fastembed embeddings, cosine similarity ≥ 0.80), fixing single-source silos caused by outlets using different headlines for the same event.
- Closed a stored-XSS vector by introducing Rust-based HTML sanitization (nh3), alongside a Django Ninja / Next.js 16 rebuild with Celery-driven ingestion, a real-time SSE breaking-news ticker, and a semantic RAG search endpoint.

Repo: github.com/emmanuelrichard01/ULTRA-NEWS

Global Rate Limiter as a Service — Vega IT Competition

- Built a distributed, cluster-safe rate limiter using an atomic Redis Lua script (token bucket via EVALSHA) so concurrent instances can never race — verified with 8 simulated nodes firing 300 concurrent requests at a 50-capacity bucket, admitting exactly 50.
- Designed a fail-open circuit breaker: on a Redis outage every instance falls back to a local in-memory bucket, self-healing via a half-open probe once Redis recovers, so traffic is never fully blocked.
- Decoupled usage logging from the hot request path (async batched writes plus a rollup job) to keep the check endpoint fast (p50 <5ms, p99 <15ms); shipped with 19 tests across 4 suites and a Docker Compose cluster demo (2 replicas + load balancer).

Modern Data Warehouse — Olist E-Commerce Analytics Platform

- Built a Dagster + dbt + DuckDB warehouse processing the complete 1.5M+ record Olist Brazilian e-commerce dataset (100% of 9 source datasets) through a Bronze–Silver–Gold Medallion architecture.
- Enforced data quality with 21 automated dbt schema tests and 9 pipeline-level checks (100% passing), backed by 80%+ pytest coverage; optimized a v2.1 release to 8x faster ingestion, ~60% faster CI, and a 20% smaller production Docker image.

Repo: github.com/emmanuelrichard01/modern-warehouse

ARCHITECTURE & SYSTEMS DESIGN

Design-stage work; not yet running in production.

CBN Data Residency Migration Reference Architecture

- Designed a reference architecture for migrating Nigerian payment transaction data to local storage ahead of the CBN's January 2027 compliance deadline (Circular PSS/DIR/PUB/CIR/001/004).

Distributed Smart Meter & Inverter Telemetry Platform

- Produced blueprint-stage architecture for a real-time IoT telemetry platform using Flink, Iceberg, and TimescaleDB.

EDUCATION

Bachelor of Engineering (B.Eng.), Computer Engineering

2021 – 2025

Caritas University, Enugu State, Nigeria — CGPA: 4.10 / 5.00

National Innovative Diploma (NID), Software Engineering

2018 – 2020

AfriHUB ICT Solutions, Abuja, Nigeria — Graduated with distinction, Top 5% of class

PROFESSIONAL CERTIFICATIONS

DataCamp Data Engineer Certification — Completed comprehensive hands-on training in modern AI & data engineering, May 2026.